

Paper II Audit -- P3: Complete-split closed form (Prop 6.2 / Cor 6.3)

Test	P3 -- Complete-split closed form (Prop 6.2 / Cor 6.3)
Gate	E
Evidence	EXACT_REPRODUCIBLE (full cover-LP) over (p,q) grid 0..12
Method	auditor's own full cover-LP vs ledger closed forms
Environment	Python 3.14.4 / networkx 3.6.1

1. Claims under test (Prop 6.2 / Cor 6.3)

$\tau_3^*(S_{\{p,q\}})$ has a two-branch closed form (non-saturated $(P+pq)/3$ for $q \leq p-1$, saturated P for $q \geq p-1$, and 0 in the degenerate cases), and $\Phi_{\tau}(S_{\{p,q\}})$ equals $p(2n+1-3p)/2$ (saturated) or $p(2n-p-1)/6$ (non-saturated). We solve the FULL cover-LP directly and compare.

2. Kill-switch (declared before running)

H3a LP == closed τ_3^* ; **H3b** Φ_{τ} closed forms == LP; **H3c** the two formulas agree at the branch boundary $q=p-1$. Falsify on any mismatch.

3. Result

Cells tested: **168**. τ_3^* mismatches: **0**. Φ_{τ} mismatches: **0**. Branch-boundary mismatches: **0**.

Verdict: VERIFIED: closed forms for $\tau_3^*(S_{\{p,q\}})$ and $\Phi_{\tau}(S_{\{p,q\}})$ match the direct full cover-LP on the whole (p,q) grid; branch boundary $q=p-1$ is consistent; degenerate cases handled.

4. Sample of the (p,q) grid

p	q	edges	$\tau_3^*_{LP}$	$\tau_3^*_{closed}$	Φ_{LP}	Φ_{closed}	branch	ok?
0	1	0	0.00	0.00	0.00	0.00	degen	y
0	2	0	0.00	0.00	0.00	0.00	degen	y
0	3	0	0.00	0.00	0.00	0.00	degen	y
0	4	0	0.00	0.00	0.00	0.00	degen	y
0	5	0	0.00	0.00	0.00	0.00	degen	y
1	0	0	0.00	0.00	0.00	0.00	degen	y
1	1	1	0.00	0.00	1.00	1.00	degen	y
1	2	2	0.00	0.00	2.00	2.00	degen	y
1	3	3	0.00	0.00	3.00	3.00	degen	y
1	4	4	0.00	0.00	4.00	4.00	degen	y
1	5	5	0.00	0.00	5.00	5.00	degen	y
2	0	1	0.00	0.00	1.00	1.00	degen	y
2	1	3	1.00	1.00	1.00	1.00	sat	y
2	2	5	1.00	1.00	3.00	3.00	sat	y
2	3	7	1.00	1.00	5.00	5.00	sat	y
2	4	9	1.00	1.00	7.00	7.00	sat	y
2	5	11	1.00	1.00	9.00	9.00	sat	y
3	0	3	1.00	1.00	1.00	1.00	nonsat	y
3	1	6	2.00	2.00	2.00	2.00	nonsat	y
3	2	9	3.00	3.00	3.00	3.00	sat	y

p	q	edges	tau3*_LP	tau3*_closed	Phi_LP	Phi_closed	branch	ok?
3	3	12	3.00	3.00	6.00	6.00	sat	y
3	4	15	3.00	3.00	9.00	9.00	sat	y
3	5	18	3.00	3.00	12.00	12.00	sat	y
4	0	6	2.00	2.00	2.00	2.00	nonsat	y
4	1	10	3.33	3.33	3.33	3.33	nonsat	y
4	2	14	4.67	4.67	4.67	4.67	nonsat	y
4	3	18	6.00	6.00	6.00	6.00	sat	y
4	4	22	6.00	6.00	10.00	10.00	sat	y
4	5	26	6.00	6.00	14.00	14.00	sat	y
5	0	10	3.33	3.33	3.33	3.33	nonsat	y
5	1	15	5.00	5.00	5.00	5.00	nonsat	y
5	2	20	6.67	6.67	6.67	6.67	nonsat	y
5	3	25	8.33	8.33	8.33	8.33	nonsat	y
5	4	30	10.00	10.00	10.00	10.00	sat	y
5	5	35	10.00	10.00	15.00	15.00	sat	y
6	0	15	5.00	5.00	5.00	5.00	nonsat	y
6	1	21	7.00	7.00	7.00	7.00	nonsat	y
6	2	27	9.00	9.00	9.00	9.00	nonsat	y
6	3	33	11.00	11.00	11.00	11.00	nonsat	y
6	4	39	13.00	13.00	13.00	13.00	nonsat	y
6	5	45	15.00	15.00	15.00	15.00	sat	y

5. Scope

The cover-LP is solved exactly (HiGHS) over all triangles/edges -- this is the FULL LP, not the symmetrized 2-variable LP of Lemma 6.1, so it also cross-validates the orbit-averaging reduction. The (p,q) grid is finite; the closed forms are algebraic in (p,q), so agreement across the grid strongly supports them. Degenerate cases $p \leq 1$ and $S_{\{2,0\}}$ (no triangles, $\tau_{3^*}=0$) are confirmed.